I Existing method for solving fully fuzzy Linear Programming Problems (FFLP) with Equality Constraints

$$\text{Max} \sum_{j=1}^{n} \tilde{c}_j \otimes \tilde{x}_j \qquad (1)$$

$$\text{st} \sum_{j=1}^{n} \tilde{a}_{ij} \otimes \tilde{x}_{ij} = \tilde{b}_i \quad \forall\, i = 1, 2, \dots, m$$

Where $\tilde{c}_j$, $\tilde{x}_j$, $\tilde{a}_{ij}$, and $\tilde{b}_j$ are non-negative triangular fuzzy numbers.

Def 2.6: a fuzzy number $\tilde{A}$ is non-negative iff $\mu_{\tilde{A}}(x) = 0 \ \forall x < 0$

Where $\mu_{\tilde{A}}(x)$ is the membership function. This takes on different forms depending on the type of fuzzy number. A typical trapezoidal fuzzy number has the general membership function

$$\mu_{\tilde{A}}(x) = \begin{cases} \frac{x-a}{b-a} & a \le x < b \\ 1 & b \le x \le c \\ \frac{x-d}{c-d} & c < x \le d \\ 0 & \text{otherwise} \end{cases} \qquad (2)$$

Refering back to (1), Lets now make some assumptions about our fuzzy numbers

Def 2.8: A trapezoidal fuzzy number $\tilde{A} = (a, b, c, d)$. It is λ-cut, denoted A^{λ}, is defined as

$$A^{\lambda} = [a + (b-a)\lambda, d - (d-c)\lambda] \ , \ 0 \le \lambda \le 1$$

Def 2.10: A trapezoidal fuzzy number $\tilde{A} = (a, b, c, d)$ is said to be non-negative iff $a \geq 0$, and is said to be non-positive iff $a \leq 0$.

Applying these definitions and fuzzy arithmetic to expand and then simplify results in a crisp LP. Note that

Remark 2.3: Let $\tilde{a}^{\lambda} = [\underline{a}(\lambda), \bar{a}(\lambda)]$ be a parametric form of an asymmetric fuzzy number $\tilde{a}$ then its nearest symmetric triangular fuzzy number is (a_0, σ), where the core "a_0" and spread "σ" of the symmetric fuzzy number can be obtained as

$$a_0 = \frac{3}{2}\int_0^1 (\bar{a}(\lambda) - \underline{a}(\lambda))(1-\lambda)\,d\lambda$$

$$\sigma = \frac{1}{2}\int_0^1 (\bar{a}(\lambda) + \underline{a}(\lambda))\,d\lambda$$

Using

Def 2.12: $\tilde{A}_1 (a_1, b_1, c_1, d_1)$ and $\tilde{A}_2 = (a_2, b_2, c_2, d_2)$ is said to be equal iff $a_1 = a_2$, $b_1 = b_2$, $c_1 = c_2$, and $d_1 = d_2$.

Then (2.6) in the book maximizes the core. If (2.6) has a unique optimal solution, then we are done. If it has alternate optimal solutions (2.7) in the book minimizes spread.

II Limitations of the Existing Method

A) Limitations

To be clear the above method can solve fully fuzzy, non-negative triangular fuzzy numbers. All the variables are fuzzy. The method cannot solve

1) FFLP with non-negative trapezoidal fuzzy numbers

2) Equality constraints where some or all coefficients are unrestricted triangular or fuzzy numbers

↳ Positive or negative

B) Shortcomings

This method needs to approximate all the coefficients into its nearest symmetric fuzzy numbers. Therefore the optimal values found may not exactly satisfy all constraints.

II Kumar et al.'s Method to find Non-negative Fuzzy optimal Solution to FFLP with equality constraints.

The following is proposed to remedy the limitations and shortcomings of the previous existing method. This method is presented to find the non-negative fuzzy optimal solution to the FFLP constraints that are

- Unrestricted coefficients
- Trapezoidal or triangular

Given

$$\text{Max/Min} \sum_{j=1}^{n} \tilde{c}_j \otimes \tilde{x}_j$$

$$\text{s.t.} \sum_{j=1}^{n} \tilde{a}_{ij} \otimes \tilde{x}_j = \tilde{b}_j \quad \forall i = 1, 2, \ldots, n$$

Where $\tilde{c}_j, \tilde{a}_{ij}, \tilde{b}_j$ are unrestricted triangular or trapezoidal fuzzy numbers and $\tilde{x}_j$ is a non-negative triangular or trapezoidal fuzzy number

A) Step 1

Let $(\tilde{a}_{ij}, \tilde{b}_{ij}, \tilde{c}_{ij}, \tilde{d}_{ij}) = \sum_{j=1}^{n} \tilde{a}_{ij} \otimes \tilde{x}_j = \tilde{b}_j \quad \forall i = 1, 2, \ldots, n$

Let $\tilde{b}_j = (b_i, g_i, h_i, k_i)$

B) Step 2.

The constraints can be rewritten as

$$\sum_{j=1}^{n} a'_{ij} = b_i \qquad \sum_{j=1}^{n} c'_{ij} = h_i \qquad \forall i = 1, 2, \ldots, n$$

$$\sum_{j=1}^{n} b'_{ij} = g_i \qquad \sum_{j=1}^{n} d'_{ij} = k_i$$

c) Step 3

Suppose that the FFLP has 'l' basic feasible solutions where the tth feasible solution is of the form

$$\{x_j^t, y_j^t, z_j^t, w_j^t\}$$

The goal is then to find the min/max out of these 'l' solutions. Now some weird theory. It was proposed that if the max/min of

$$\max/\min \mathbb{R}\left(\sum_{j=1}^{n}(p_j, q_j, r_j, s_j) \otimes (x_j^t, y_j^t, z_j^t, w_j^t)\right)$$

is

$$\mathbb{R}\left(\sum_{j=1}^{n}(p_j, q_j, r_j, s_j) \otimes (x_j^0, y_j^0, z_j^0, w_j^0)\right) \qquad (3).$$

then

$$\max/\min \sum_{j=1}^{n}(p_j, q_j, r_j, s_j) \otimes (x_j^t, y_j^t, z_j^t, w_j^t)$$

is in fact (3) as well! Well, what is $\mathbb{R}$? It is defined as

$$\mathbb{R}(a, b, c, d) = \frac{1}{4}(a+b+c+d)$$

Where $\mathbb{R}$ is the Takagi-Sugeno-Kang (TSK) method of defuzzification. Generally it is written

$$\frac{\sum_i (\mu_i Y_i)}{\sum_i \mu_i}$$

μ_i : Membership value

Y_i : Fuzzy value.

In our case $\mu_i = 1 \ \forall \ i = 1, 2, 3, \ldots$

D) Step 4

Using $\sum_{j=1}^{n} (p_j, q_j, r_j, s_j) \otimes (x_j, y_j, z_j, w_j) = (p_j', q_j', r_j', s_j')$

and the definition of TR, we can rewrite the system as

$$\min/\max \frac{1}{4}(p_j', q_j', r_j', s_j')$$

s.t.

$$\sum_{j=1}^{n} a_{ij}' = b_i \qquad \sum_{j=1}^{n} c_{ij} = h_i$$

$$\forall i = 1, 2, \dots, m$$

$$\sum_{j=1}^{n} b_{ij}' = g_i \qquad \sum_{j=1}^{n} d_{ij} = k_i$$

$$x_j \geq 0, \quad y_j - x_j \geq 0, \quad z_j - y_j \geq 0, \quad w_j - z_j \geq 0$$

$$\forall j = 1, 2, \dots, m$$

E) Step 5

Solve the crisp LP above to obtain an optimal solution

$$\{x_j^*, y_j^*, z_j^*, w_j^*\}$$

F) Step 6

Find the fuzzy optimal solution of the FFLP from step 1 by substituting results from step 5.

E) Step 7

Plug in the new optimal fuzzy solution, x_j^*, back into

$$\sum_{j=1}^{n} \tilde{c}_j \otimes x_j^*$$

III Advantages of Kumar et. all's Method

A) Easy to implement

B) More accurate results. In the example given in the book it is exact.

C) Can be used to find non-negative fuzzy optimal solution of all fuzzy LP problems.